Airbnb Price Transparency A/B Test

Evaluating the Impact of Price Transparency on Booking Conversion in New Orleans

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Business Problem & Objective

Business Problem:

Users initially see a pre-tax price when browsing listings, with the full price (including fees and taxes) revealed later during checkout. This may cause:

- Price surprise
- Increased friction
- Checkout abandonment
- Reduced trust

Objective: Evaluate whether showing the full price upfront improves checkout completion rate.

Primary Metric:

$$P(Complete \mid Start\ Checkout)$$

Hypothesis

Null Hypothesis:

- Showing the full price upfront does not change checkout completion compared to the pre-tax price.

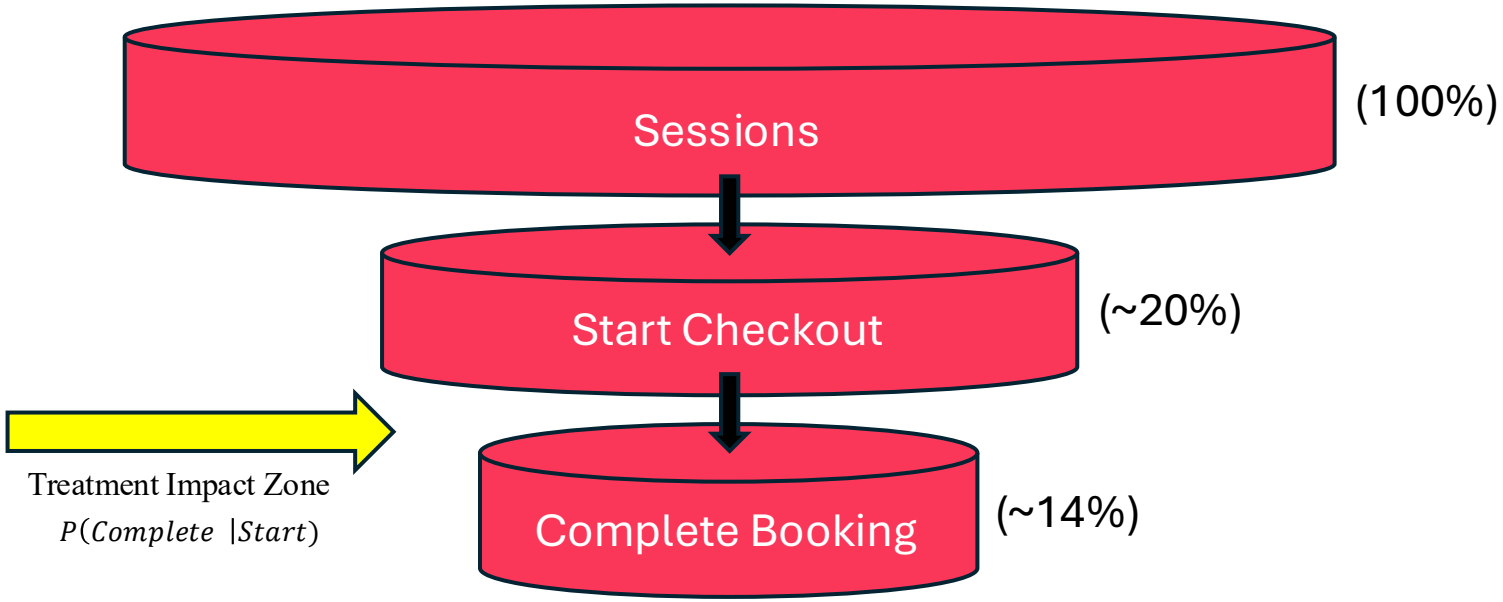
$$P(Complete\,|Start)_{Full\ Price} \leq P(Complete\,|Start)_{Pre-Tax\ Price}$$

Alternate Hypothesis:

- Showing the full price upfront increases checkout completion compared to showing the pre-tax price.

$$P(Complete\,|Start)_{Full\ Price} > P(Complete\,|Start)_{Pre-Tax\ Price}$$

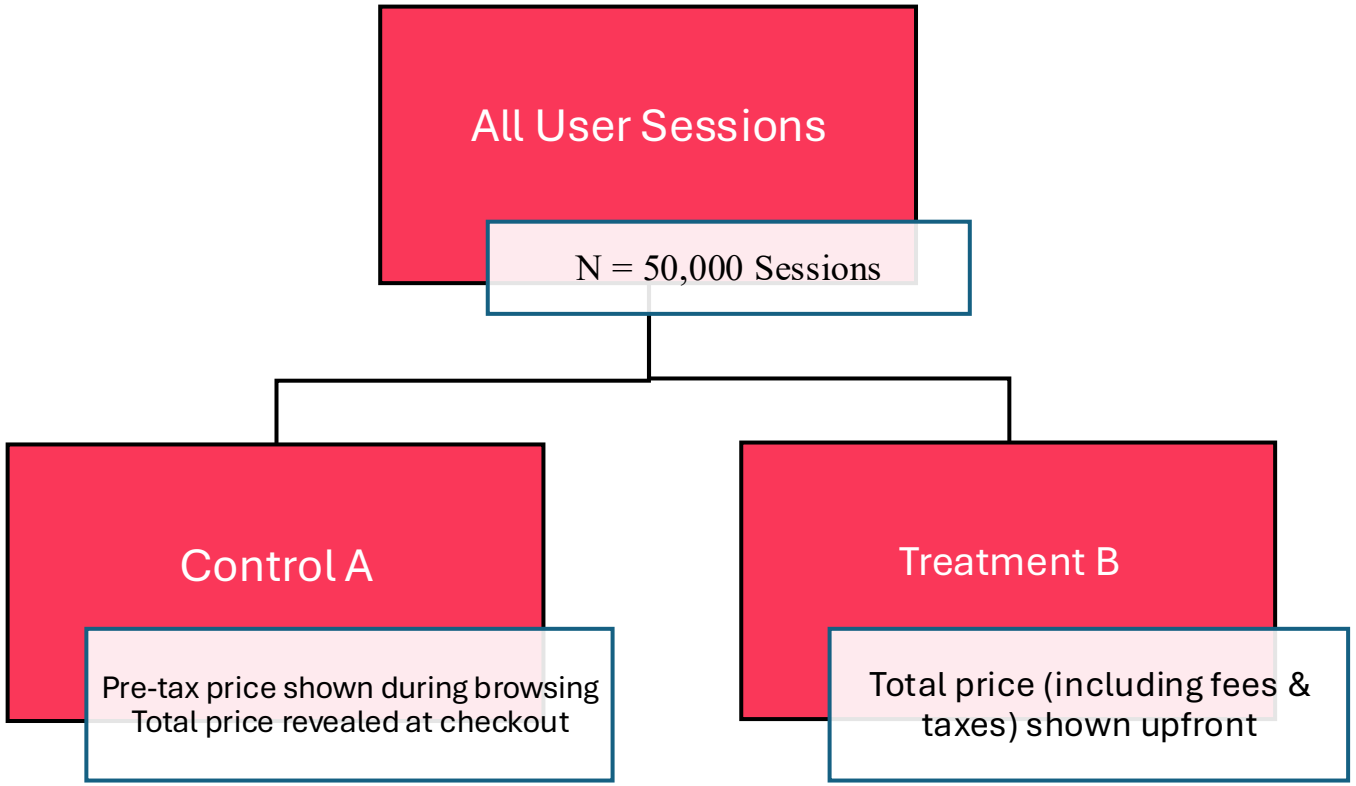
Booking Funnel & Conversion Flow



Conversion Relationship:

$$P(Booking) = P(Start)\ x\ P(Complete\ |Start)$$

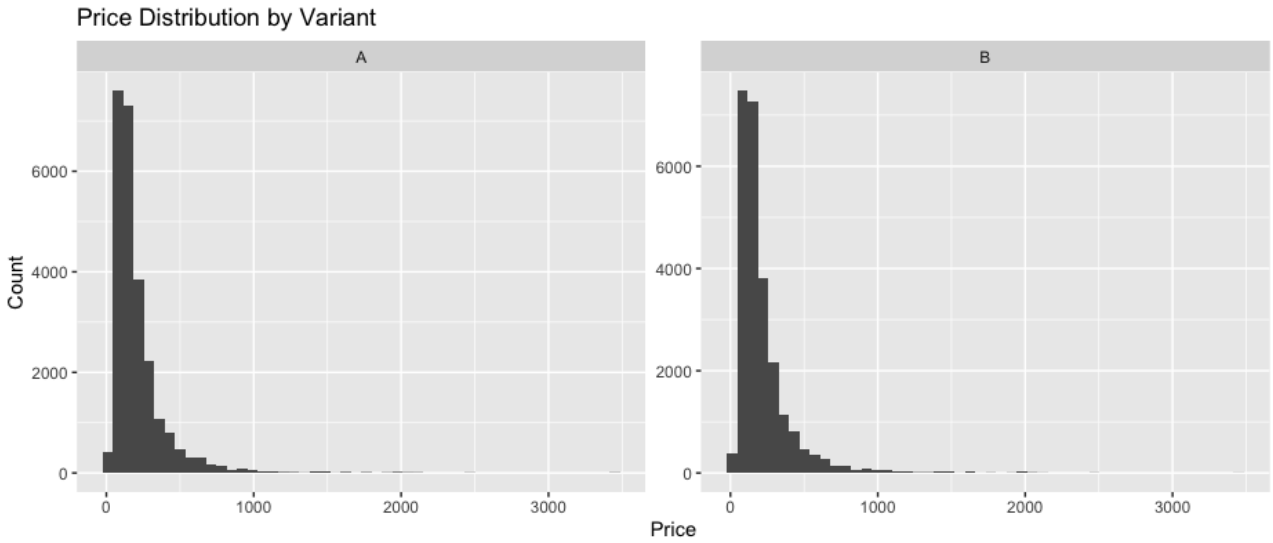
Experimental Design



Randomization & Experiment Integrity

Sample Split

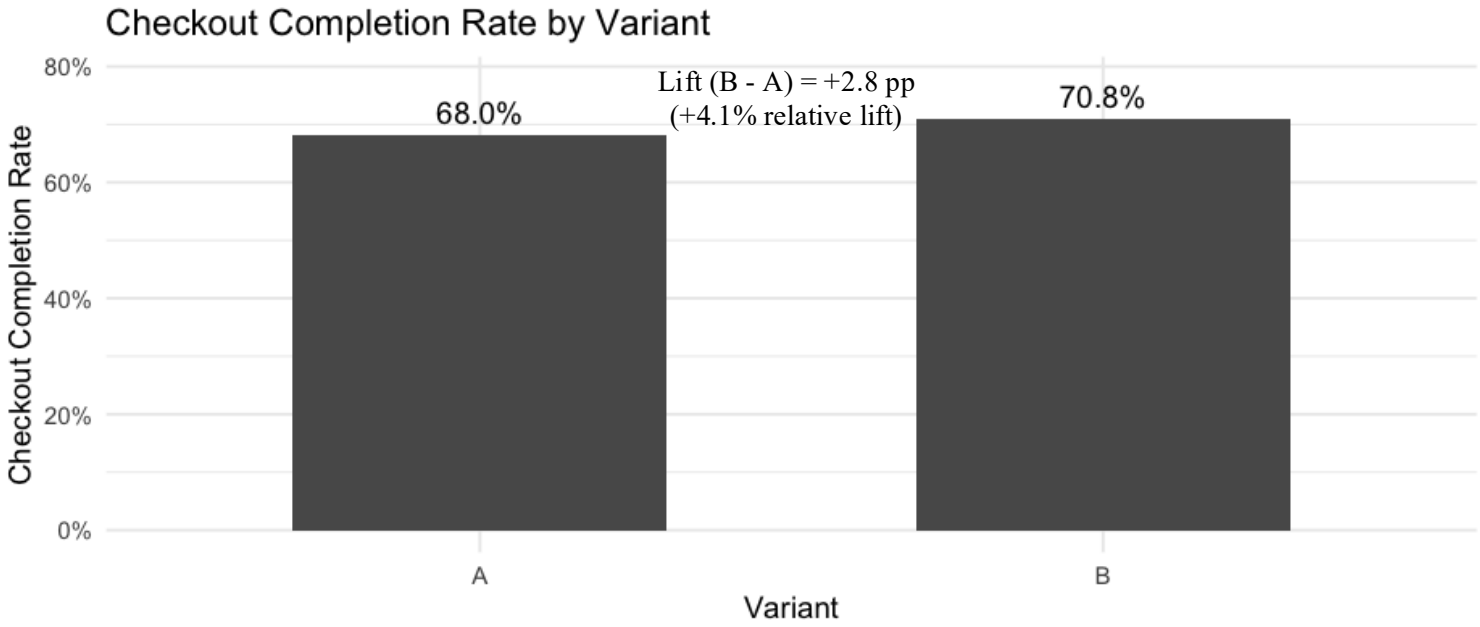
- Control (A): 25,093 sessions
- Treatment (B): 24,907 sessions
- Chi-Squared test p-value = 0.405



The even session split and similar price distributions indicate successful randomization and support a valid comparison between Control and Treatment groups.

A/B Test Results

Primary Metric: $P(Complete \mid Start\ Checkout)$



Secondary Impact: Overall Booking Conversion

Control (A): 13.7%

Treatment (B): 14.5%

Absolute Lift: +0.8 percentage points

(+5.8% relative lift)

Treatment Validation

Primary Metric: Checkout Completion Rate

Test: One-sided two-proportion z-test

p-value: 0.0012

Secondary Impact: Overall Booking Conversion

Test: One-sided two-proportion z-test

p-value: 0.0075

Conclusion

We reject the null hypothesis. Full price transparency improves booking performance.

Regression Confirmation

We run a multivariable logistic regression to confirm that the treatment effect remains significant after controlling for listing characteristics.

Model

Outcome: Completed Booking

Controls: Price, Rating, Super-host Status, Instant Book

Key Results & Interpretation

After adjusting for listing characteristics, showing full price upfront significantly increases checkout completion ($p = 0.002$). The effect remains statistically significant, confirming the observed lift is not driven by listing-level differences.

Price transparency independently drives higher checkout completion.

Business Recommendations

Recommendation: Standardize full price transparency across the platform.

Expected Impact:

- +0.8 percentage point increase in overall booking conversion
- ~5.8% relative lift in bookings
- Improved user trust and pricing transparency
- Incremental revenue growth at scale

Even modest conversion gains compound meaningfully at platform scale.

Risk Assessment

- No negative impact on booking behavior detected
- Lift remains significant after controlling for listing characteristics.